Qishu (Jack) Qin

qishuqin26@gmail.com | LinkedIn: www.linkedin.com/in/qishuqin | GitHub: JackQinPSU (Qishu(Jack) Qin)

EDUCATION

The Pennsylvania State University — University Park, PA
College of Engineering — *Expected Graduation Month:* ***May 2028***
Bachelor of Science in Computer Science — GPA: **3.8/4.00**
Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Functional Programming, Algorithm Analysis, Linear Algebra, Multivariable Calculus, Systems Programming, Discrete Math

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, Java, SQL, C++, C, Bash
- Frameworks/Libraries: React, Node.js, PyTorch, scikit-learn, Pandas
- Tools: Linux, Docker, Git, AWS (S3, ECS, Lambda, DynamoDB, CloudWatch), Kafka, PostgreSQL, MySQL

WORK EXPERIENCE

Phigent Robotics — *May 2026 – August 2026*
Data Platform Engineering Intern — *Shanghai, China*

- Built and migrated **10+** batch data workflows on an internal platform orchestrating autonomous-driving perception modules, writing Python pipeline code for data fetch, per-clip isolation, subprocess execution, validation, and upload.
- Composed multi-stage DAG templates of up to **8 stages**, configured concurrency, queues, and failure handling, and operated jobs spanning **1000+ clips** — monitoring logs, rerunning failures, and validating outputs.
- Cut upstream dataset fetches **50% per clip** in the 3D traffic-light workflow by removing a redundant local-map dependency, reducing object-storage reads, temporary-disk usage, and dependency-related failures.
- Migrated legacy perception modules across company infrastructures: rebuilt Docker images, reconfigured registries, mirrors, and environment variables, and resolved dependency and runtime failures on shared GPU cluster queues.

Pennsylvania State University — *April 2026 – Present*
Undergraduate Researcher — *State College, PA*

- Integrated a native C++ LinUCB contextual bandit into CaDiCaL's restart system and instrumented clause quality, conflicts, restart frequency, and PAR-2 to support controlled baseline and RL ablations across SAT benchmark.
- Traced a **45% runtime regression** on an RL-active instance after **12K+ policy updates** to policy and logging overhead, redirecting evaluation toward harder datasets and lower-overhead reward collection.

COSCO Shipping (North America) Inc. — *June 2025 – August 2025*
Operational Data Intern — *Secaucus, NJ*

- Streamlined rental data workflows across teams, reducing reporting errors by **~30%.**
- Collaborated with PM to improve internal tracking system, increasing data consistency across departments.

TECHNICAL PROJECTS

High-Frequency Trading Order Book Simulator (C++)

- Built a low-latency C++17 limit order book engine implementing price-time priority with fixed-point pricing and **O(1)** order cancellation for efficient and deterministic order management under high-frequency workloads.
- Benchmarked the engine on 1M simulated orders, sustaining **1.1M+** orders per second with approximately **873 ns** average processing latency per order, validating correctness through stress testing.

Formless – AI-Powered Benefits Application System (React, TypeScript, Gemini, pdf-lib)

- Built an LLM-driven system that extracts structured data from natural language and dynamically collects missing inputs, compressing multi-hour government benefit applications into a guided adaptive workflow.
- Selected as a **Top 10** North America Finalist in the 2026 Google Developer Groups on Campus Solution Challenge.

LEADERSHIP & AWARDS

Optiver FutureFocus Program | Selected Participant

- Completed a five-day trading-technology program with hands-on Python/OOP, system-design and risk-control.

AWS Student Builder's Club at Penn State | Technical Chair

- Build and maintain the club website; lead hands-on workshops on AWS fundamentals and cloud projects.

C0mpiled-6 Hackathon | Winner of lyra Communication Track

- Built SyncStudy, an AI group-study optimizer using live feedback to recommend the next topic.